

PharmaGuard — Weekly Progress Update #2

Multi-Scale Benchmark Expansion (165 Pairs), Local LLM Integration, Conference Paper Manuscript, and Interactive Dashboard

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1. Feedback & Guidance from Previous Meeting (Recap)

In our previous meeting and update, we established the following key directions:

- 1. **Scoring-Inert Clinical Context:** Following our 40-pair test of the indication concordance layer (which yielded a low-power null result of 4/40 triggers without metric movement), we agreed to maintain the disease-context layer as an **explicit, scoring-inert provenance flag** on reports rather than applying ungrounded numerical scalar penalties.
- 2. **Scale & Rigorous Benchmarking:** You advised moving beyond small-sample toy evaluation (15 pairs) to test whether PharmaGuard's triage gates and tri-source fusion hold up under large-scale, real-world postmarketing surveillance conditions.
- 3. **Conference Paper Drafting:** We received the go-ahead to transition our literature review, methodology, and empirical findings into a formal academic manuscript targeting IEEE / ACM / JAMIA health informatics venues.

This document summarizes our major progress across all three directives.

2. Multi-Scale Benchmark Expansion (165 Total Pairs Evaluated)

To rigorously stress-test the system, we expanded evaluation from our original 15-pair prototype to **three stratified benchmark cohorts totaling 165 curated drug–event pairs** (with 151 distinct searchable presets):

Benchmark Cohort	Cohort Size (\$N\$)	Cohort Composition & Focus	Key Findings & Metrics
1. Core Showcase Cohort	15 Pairs	7 confirmed positive toxicities + 8 negative controls. Frozen Sprint 3 baseline.	Strict \$F_1 = 0.9231\$, Lenient \$F_1 = 0.9333\$. Halves clinician over-caution rate from 25.0% down to 12.5% vs. ungrounded LLM.
2. Top Prescribed Blockbusters	50 Pairs	25 high-recognition, high-mortality FDA Boxed Warnings vs. 25 negative controls across Statins, ACEi/ARBs, Antibiotics, Antidepressants, Opioids, and Anticonvulsants.	Lenient \$F_1 = 0.9388\$, Recall = 92.0% (23/25 Boxed Warnings caught), Specificity = 96.0% (24/25 cleared), Strict Precision = 1.0000 (10/10).

Benchmark Cohort	Cohort Size (\$N\$)	Cohort Composition & Focus	Key Findings & Metrics
3. OMOP Expanded Reference Standard	100 Pairs	50 positive vs. 50 negative controls from the OHDSI OMOP gold standard across 4 acute organ failure phenotypes: Acute Liver Injury, Acute Renal Failure, Myocardial Infarction, Upper GI Bleeding.	Strict Specificity = 1.0000 (50/50 negative controls cleared; zero false alarms), Lenient Specificity = 0.9800 (49/50), Lenient Precision = 0.9333 (14/15).

Key Scientific Discovery: "PRR Denominator Dilution"

While evaluating the 100-pair OMOP reference standard, we made an important empirical observation that directly explains a major challenge in modern pharmacovigilance:

- **The Observation:** On chronic, high-utilization medications (such as long-term statins or PPIs), raw disproportionality metrics like PRR often drop below the classical Evans threshold of \$2.0\$, despite significant clinical literature and confirmed biological mechanisms.
- **Why this happens:** When tens of millions of patients take a drug continuously, the sheer volume of incidental background reports inflates the denominator of the 2x2 contingency table, artificially depressing the PRR statistic.
- **Why this matters for PharmaGuard:** A purely statistical tool will drop these real signals. However, PharmaGuard's **tri-source fusion** uses ChEMBL receptor pharmacology and PubMed literature grading to prevent them from being discarded, routing them safely to **MONITOR** rather than **DO_NOT_ESCALATE**. This provides strong empirical justification for our dual-metric (Strict vs. Lenient) evaluation framework in the paper.

3. Zero-Cost, Unmetered Local LLM Integration (Ollama **qwen2.5:7b**)

Executing 165 multi-source pipeline runs against commercial cloud APIs (like Google Gemini Flash) introduced severe rate-limiting bottlenecks (e.g., 15 RPM free-tier ceilings) and recurring API costs.

What we built:

- We integrated the **Ollama local inference daemon (qwen2.5:7b)** directly into PharmaGuard's LLM factory layer.
- Local LLM inference runs 100% on-device via **http://localhost:11434**, with unmetered throughput, zero API fees, and complete offline privacy.
- We evaluated the entire 50-pair Blockbuster benchmark and 100-pair OMOP benchmark completely locally through Ollama, achieving identical deterministic schema extraction without dropping a single call.
- The system retains seamless dual-backend support: users can switch between local Ollama and Cloud Gemini Flash via a single configuration toggle.

4. Complete Academic Conference Paper Manuscript (Drafted)

We have completed the full first-draft manuscript of our conference paper, targeted for submission to **IEEE BIBM (International Conference on Bioinformatics and Biomedicine)**, **ACM CHIL (Conference on Health, Inference, and Learning)**, or **JAMIA**:

- **Location:** `docs/paper/PharmaGuard_Conference_Paper.md` (336 lines, ~8,500 words).
- **Structure (9 Formal Sections):**
 1. **Title & Abstract:** Formulates the postmarketing signal triage problem, triage bottleneck, and summary of 165-pair empirical results.
 2. **Introduction:** Contrasts spontaneous reporting realities against pre-approval RCT limitations; details the failure modes of ungrounded LLMs (hallucinated confidence, historical regulatory confusion, parametric leakage).
 3. **Related Work:** Thoroughly positions PharmaGuard against Evans PRR/ROR statistical baselines, modern agentic architectures (ReAct, PSEBench, DruGagent), and EHR-dependent causal AI (Toonsi et al.), highlighting our public-API accessibility advantage.
 4. **System Architecture & Methodology:** Mathematical definitions of the closed-form confidence fusion formula, Gate 1 empirical safety stop, ChEMBL MoA retrieval, PubMed clinical rubric v1.0, and the scoring-inert ATC indication isolation wall.
 5. **Experimental Setup:** Master table of all three benchmark cohorts (\$N=165\$), ground-truth curation standards, and the dual-metric evaluation framework (Strict vs. Lenient).
 6. **Results:** Comprehensive tables reporting TP, FP, TN, FN, Precision, Recall, Specificity, and \$F_1\$ with exact **Wilson 95% Confidence Intervals** across all three cohorts.
 7. **Clinical Case Studies:** In-depth case walkthroughs on *Montelukast* (suicidal ideation), *Metformin* (lactic acidosis vs. hypoglycaemia polypharmacy), *Lisinopril* (cough), and *Albuterol* (paradoxical bronchospasm).
 8. **Discussion & Limitations:** Analysis of PRR denominator dilution, the MARCH adversarial critic anti-leakage audit, and current boundary conditions.
 9. **Conclusion & Reproducibility:** Availability statement for code, datasets, Docker deployment, and Streamlit evaluation dashboard.

5. Evaluation Dashboard Multi-Cohort Integration & Live Signal Triage

We significantly upgraded the Streamlit evaluation dashboard (`scripts/dashboard.py` / `dashboard/app.py` running at `http://localhost:8501`):

1. Global Multi-Cohort Switcher:

- Added a top-bar segmented control switcher allowing reviewers to dynamically toggle the dashboard across:
 - **Core Showcase [15]**
 - **Top Prescribed [50]**
 - **OMOP Expanded [100]**
- Switching the cohort instantly re-renders the Overview KPI cards, Confusion Matrices, Metric Breakdown tables, and Per-Pair evidence drill-downs.

2. Searchable Live Signal Triage Playground:

- Implemented a dedicated Live Triage tab loaded with **151 searchable benchmark presets** alongside support for arbitrary custom drug–event entry.

- Executes live, real-time multi-source data extraction across openFDA FAERS, ChEMBL MoA targets, PubMed abstracts, and local Ollama inference with live step-by-step progress tracking.

3. Methodology Probes & Auditing:

- Displays the Adversarial Critic probe audit table (MARCH framework pattern) showing 4/4 detection of parametric training-data leakage.
- Visualizes side-by-side evidence waterfalls for polypharmacy confounding de-biasing.

4. Engineering Fixes & UI Polish:

- Replaced raw emojis with official Google Material Symbols across all tabs and badges.
- Fixed a `NoneType` caching exception in `ChEMBLTool` and `PubMedTool` when disk caching is toggled off.
- Added automatic module hot-reloading (`importlib.reload`) in `live_triage.py` to prevent persistent Streamlit memory from retaining stale bytecode.
- Verified that all **242 / 242 unit tests** pass with 100% green status via pytest.

6. Capstone Presentation Deck Synchronization

We fully synchronized the mid/end-semester capstone presentation materials with the 165-pair benchmark scale:

- **Slide Outline** (`docs/presentation/End_Semester_Slide_Content.md`): Updated the 18-slide presentation deck to reflect the multi-cohort evaluation, PRR denominator dilution findings, local Ollama integration, and live dashboard demo flow.
- **Delivery Notes** (`docs/presentation/SLIDE_DECK_NOTES.md`): Updated presenter cue cards, timing checkpoints (15-minute budget), and anticipated defense/viva defense questions (e.g., explaining why OMOP strict recall is lower due to chronic utilization, and defending why the indication concordance layer is scoring-inert).

7. Summary of Completed Milestones

Target Area	Status	Deliverables / Evidence
Benchmark Scaling	COMPLETE	165 total drug–event pairs evaluated across 3 stratified cohorts (15 Core, 50 Blockbusters, 100 OMOP).
Local LLM Backend	COMPLETE	Zero-cost Ollama <code>qwen2.5:7b</code> integration; fully tested and reproducible offline.
Academic Manuscript	COMPLETE (DRAFT)	9-section conference paper manuscript at <code>docs/paper/PharmaGuard_Conference_Paper.md</code> .
Interactive Dashboard	COMPLETE & LIVE	Multi-cohort switcher, 151 live triage presets, and methodology probes running on <code>localhost:8501</code> .
Defense Slide Deck	COMPLETE	18-slide end-semester presentation deck and speaker notes synchronized with 165-pair data.

Target Area	Status	Deliverables / Evidence
Code Quality & Tests	COMPLETE	242/242 pytest unit tests passing cleanly; clean git tree on origin/main .

8. Discussion Points & Next Steps for Your Guidance

We would appreciate your feedback and advice on the following points:

1. **Target Publication Venue:**
 - The manuscript is currently budgeted for a 6–8 page double-column conference format (ideal for **IEEE BIBM 2026** or **ACM CHIL 2026**) or can be extended into a full journal paper for **JAMIA (Journal of the American Medical Informatics Association)**. What venue would you prefer us to format the submission for first?
2. **Manuscript Review:**
 - Would you like to review the markdown manuscript directly ([docs/paper/PharmaGuard_Conference_Paper.md](#)), or should we compile it into the formal IEEE/ACM LaTeX template with embedded vector PDF figures for your review?
3. **Capstone Defense Preparation:**
 - For our upcoming capstone presentation, would you like us to conduct a 10-minute dry-run presentation and live dashboard demonstration during our next meeting?